Hyperbola-Band Attention Supervision for Detection in GPR B-Scan Images

Ming Li, Christopher Gomez, Shusuke Miyata, and Norifumi Hotta

Abstract—In ground penetrating radar (GPR) B-scan images, subsurface targets appear as hyperbolic reflections, and their accurate detection is a prerequisite for target localization and parameter inversion. Existing deep learning detectors typically adopt axis-aligned bounding boxes as the annotation format, overlooking the strip-like, symmetric, and parametric geometry of hyperbolic echoes. Moreover, the purely data-driven nature of existing attention mechanisms makes attention maps prone to drift under few-shot conditions, deviating from true hyperbolic structures and being misled by background clutter, which ultimately undermines detection stability. To address these issues, this paper proposes a hyperbola-band attention supervision method. We develop the hyperbola-band annotation tool (HBAT), a dedicated annotation tool that represents each target as a parametric band defined by its vertex, slope, span, and thickness. From these parameters, HBAT simultaneously rasterizes a band mask and derives the corresponding bounding box, ensuring geometric consistency between the two label types. Building on these annotations, an attention branch in the network learns to focus on hyperbola regions under the explicit supervision of this mask, and the resulting spatial prior is injected into the detection features, with the whole network trained end to end. Experiments on a public GPR dataset show that the proposed method consistently outperforms the baseline across training-data ratios, with the largest gain raising the F1-score from 0.68 to 0.84, while substantially reducing variance across random seeds. The method also surpasses four representative data-driven attention modules, demonstrating the effectiveness of explicit geometric supervision for hyperbola detection in GPR B-scans, with consistent gains confirmed on an additional GPR dataset..

Index Terms—Ground penetrating radar (GPR), hyperbola detection, attention supervision, deep learning.

I. Introduction

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As a widely recognized non-destructive inspection technique, Ground Penetrating Radar (GPR) provides high-resolution, near-real-time imaging of internal structures by employing high-frequency electromagnetic waves, allowing researchers and engineers to visualize features embedded within otherwise opaque media [1]. GPR radargram interpretation thus extends well beyond conventional geological surveys, constituting a key technology underlying a wide range of applications, including infrastructure maintenance, archaeological interpretation, military operations [2], and driftwood monitoring in river systems [3].

In GPR B-scan images, hyperbolic reflection patterns are a hallmark of discrete subsurface objects such as pipes, boulders, and driftwood. The shape of these hyperbolas encodes depth and size. Accurately identifying these hyperbolas is therefore essential for reliable target detection and parameter inversion [4].

Early hyperbola detection methods relied on handcrafted features and analytical curve fitting [5], but their performance is highly sensitive to manually designed priors and degrades significantly under heterogeneous subsurface conditions [6]. Deep learning has since become the dominant paradigm [7], [8], yet its effectiveness relies on large-scale, well-annotated training data, rarely available in practical GPR surveys due to the high cost of data collection and expert annotation. Moreover, standard detectors rely on rectangular bounding boxes that only coarsely delineate the spatial extent of an echo, without encoding fine-grained geometric properties of hyperbolas, such as vertex position, curvature, and band thickness. This structural information loss limits the network's ability to learn target-specific cues, and becomes especially detrimental under few-shot conditions, where limited training samples leave the model with no means to compensate for the missing geometric guidance, resulting in severe overfitting and unstable performance.

To steer the model toward real hyperbola structures and suppress background clutter,attention mechanisms have been introduced to help models focus on target regions [9]–[14]. They reweight features along the channel or spatial dimensions. These mechanisms are typically implicit and data-driven, and learning where to attend is itself an additional burden on top of the primary detection task [15]. Under few-shot conditions, the network has too little data to calibrate both the detection backbone and the attention weights [16], so attention maps tend to deviate from the actual hyperbolic structures and become diluted by background clutter.

To address the aforementioned limitations, this paper proposes a hyperbola-band attention supervision method. On the annotation side, the geometric shape of each hyperbola is encoded with a small set of parameters; on the learning side, this geometric information is converted into supervisory signals for an attention branch to boost detection performance.

The main contributions of this work are summarized as follows:

1) A hyperbola-band annotation tool (HBAT) is developed for labeling hyperbola-band targets in GPR images. Each target is parameterized by a compact set of interpretable attributes: apex position, curvature (slope), lateral span, and band thickness, supporting interactive adjustment during annotation. The tool exports both shape parameters and derived rectangular bounding boxes simultaneously, eliminating the need for separate manual labeling of the two representations.
2) An explicit attention supervision mechanism based on geometric priors is proposed. Hyperbola-band masks generated from parametric annotations directly constrain the attention distribution, injecting explicit spatial priors of targets into the detection network and improving detection performance via end-to-end training.
3) Experiments on a public GPR dataset of limited scale (318 samples in total) show that the proposed method improves the F1-score from 0.68 to 0.84 at a training ratio of 0.7, with a notable enhancement in model stability. It also outperforms multiple mainstream attention modules, which validates the effectiveness of the proposed method.

II. Related work

A. GPR Hyperbola Detection Methodologies

Fig. 1 shows the GPR data acquisition process: a transmitter-receiver antenna pair traverses along the scanning path, with measurement points set at regularly spaced time instants $T_0, T_1, T_2, \ldots$ (or, alternatively, at uniformly spaced spatial intervals). At each measurement point, an electromagnetic pulse emitted by the transmitting antenna is reflected at successive dielectric boundaries [17], such as the air-soil and soil-buried object interfaces. A sequence of reflections is then recorded by the receiving antenna, yielding an A-scan trace. Repeating this process at successive measurement points and arranging the resulting A-scans along the survey line produces a two-dimensional radargram, referred to as a B-scan. Directly above the buried target, the wave travels the shortest path between transmitter and receiver, giving the minimum travel time. Moving the survey point away from this position lengthens the travel time. This progressive increase in travel time produces the familiar hyperbolic signature seen in GPR profiles.

Despite this well-established principle, robust hyperbola detection remains highly challenging in practice, owing to weak reflections from deeply buried or small targets, strong clutter and noise, and hyperbola deformation or overlap.

Early automated approaches relied on hand-crafted features and analytical fitting, screening candidates with neural preprocessing, Viola–Jones, or histogram of oriented gradients (HOG) features with a support vector machine (SVM) classifier, and then localizing them via the Hough transform or template/least-squares fitting [18]–[21]. Dou et al. proposed a real-time framework that combines thresholding, column-connection clustering (C3), a lightweight two-feature model, and orthogonal-distance fitting. This framework is designed to handle hyperbolas that are crossing, distorted, or incomplete [4]. Such pipelines work in constrained settings but remain computationally expensive, sensitive to parameter discretization, and dependent on prior template design or user intervention [21]–[24].

Deep learning shifted the field from sliding-window recognition toward end-to-end detection on full B-scans. Among two-stage detectors, the Faster region-based convolutional neural network (Faster R-CNN) was first trained on simulated radargrams to offset scarce field labels [23]. Lei et al. subsequently coupled Faster R-CNN with double cluster seeking estimate (DCSE) clustering and column-based transverse filter points (CTFP) extraction, which fit the hyperbola after detection to localize buried objects [25]. Although accurate, the proposal-then-refine design of such two-stage pipelines makes inference comparatively slow, so one-stage detectors were favored for real-time deployment. Liu et al. used the single shot multibox detector (SSD) to identify hyperbolic regions of rebar, combining migration and binarization to estimate rebar position and depth [26], and later applied You Only Look Once version 3

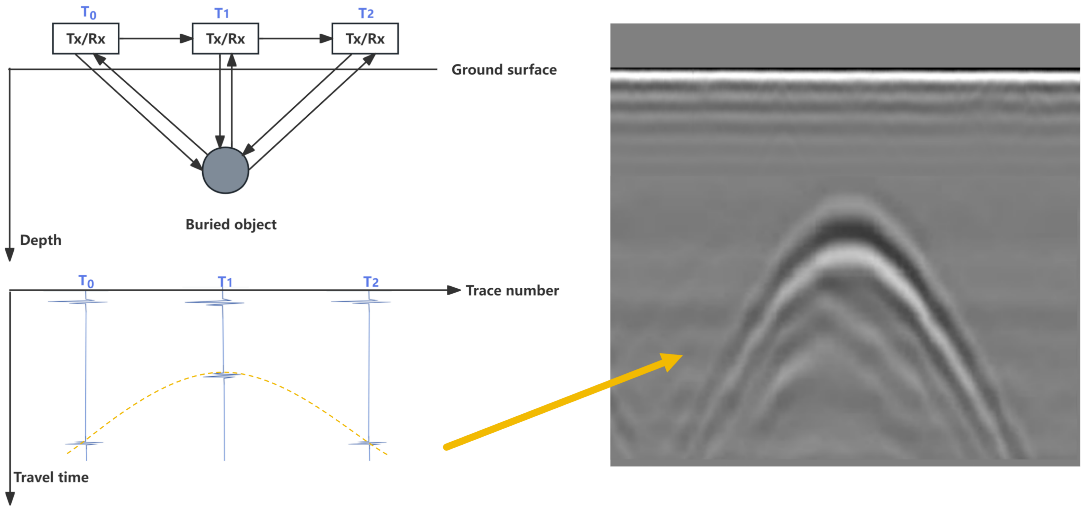


Fig. 1. Formation of a GPR B-scan.

(YOLOv3) with iterative thresholding to pipeline detection, achieving approximately 12 fps with depth estimation errors within 0.04 m [27]. Subsequent YOLO variants were coupled with explicit curve fitting: Sun et al. combined YOLOv5s with three-point circle fitting for root localization and diameter estimation [28], and Zhu et al. integrated YOLOv7 with keypoint annotation and two-stage curve fitting to invert pipeline burial depth and radius [29]. YOLO-based detectors excelled in speed, but their local features and non-maximum suppression (NMS) still struggled with crossing and incomplete hyperbolas, motivating a shift toward transformer-based, end-to-end approaches. Jin et al.'s GPR-Former used a transformer to directly regress hyperbola parameters, refined via a symmetry-constrained analytical solution, outperforming both C3 and transfer-learning-based methods in precision and recall [30].

The field of GPR hyperbola detection has thus seen a shift toward more powerful detection networks and a broader push from 2D applications toward 3D reconstruction and multi-sensor fusion [31], [32]. Across all these paradigms, however, the hyperbola is still annotated and predicted as an axis-aligned bounding box: the parametric geometry of the echo enters the pipeline only through a separate post-hoc curve-fitting stage, and detection performance degrades markedly when labeled data are scarce. How to exploit the band-like geometry of the hyperbola during detector training itself, rather than recovering it after detection, remains largely unexplored and motivates the method proposed in this work.

B. Attention Mechanisms in GPR Hyperbola Interpretation

Attention mechanisms were originally introduced in general image recognition to let a network emphasize informative regions or channels instead of treating all spatial positions and feature channels uniformly: a lightweight sub-network learns a set of weights from the feature map itself, and these weights rescale the original features so that informative regions or channels are amplified while less relevant ones are suppressed.

In GPR interpretation, this idea has been adapted so that models respond to hyperbola-relevant structures rather than processing the whole radargram uniformly. Wang et al. first use one network to suppress the direct-wave interference that dominates the upper portion of every B-scan; a second network, built around GPR domain knowledge as prior information, then applies attention together with a feature-fusion step to help the network concentrate on the remaining hyperbola, before it is fitted with a least-squares curve and localized through geometric relations [33].

A second line of work instead exploits the fact that the two arms of a hyperbola have a large spatial extent, spreading across many neighbouring measurement positions and the full recording time window – so that evidence located far apart in the image can still be relevant to identifying the same target. Transformer-style self-attention is naturally suited to this setting, since it lets any two positions in the image establish a direct connection rather than being confined, as convolution is, to a local neighbourhood. GPR-Former adopts this form of attention to regress the hyperbola's geometric parameters, such as its apex position and curvature, directly from the B-scan [30].

A third line of work goes further still, embedding the hyperbola's geometry directly into the structure of the attention module itself. To detect the curved, multiply-reflected pattern produced by rebar in concrete, Li et al. let the attention module adapt its receptive field to the target's expected curvature before region proposals are generated, so that the earliest processing stages are already biased toward hyperbola-like structures [34].

Despite this progression toward stronger geometric assumptions, all of these methods share the same limitation: prior knowledge shapes only the network's input or architecture, while the attention weights themselves are still learned indirectly, from a loss defined on the final output. This matters most under the few-shot conditions common in GPR surveys, where the attention map can still drift toward clutter even when the surrounding architecture is built around the target's shape. We take a different approach: we derive an explicit attention target directly from the parametric annotation and use it to supervise the attention branch, imposing the hyperbola's spatial prior on attention directly rather than relying on the architecture alone.

C. Annotation of hyperbolic reflections

Training a detection model requires large quantities of manually annotated images, and a variety of general-purpose annotation tools have been developed to meet this need. Early systems such as LabelMe enabled collaborative image labelling at scale [35], and the annotation pipelines behind large-scale benchmarks such as Microsoft common objects in context (MS-COCO) further standardized this process [36]. Building on these efforts, widely used open-source tools such as LabelImg [37] and the computer vision annotation tool (CVAT) [38] offer interactive interfaces for a common set of geometric shapes: axis-aligned rectangles, polygons, key points, and pixel-wise masks. Being generic and domain-agnostic, these shapes are precisely what makes such tools broadly applicable across natural-image tasks.

In GPR interpretation, hyperbolic reflections are typically annotated with these same general-purpose shapes. Bounding boxes remain the most common choice for subsurface targets in B-scans [7], whereas a growing body of work instead marks the apex of each hyperbola as a keypoint and treats it as the localization target [14], [39]; segmentation masks have also been used to delineate the hyperbola or its apex region more tightly [40]. Each of these schemes, however, captures only part of the geometry: a bounding box records merely the extent of the echo, a single keypoint pins down only the apex while discarding the opening and curvature of the curve, and segmentation masks demand dense per-instance labelling yet still do not expose the underlying hyperbola parameters. None of them aligns with the parametric, band-shaped nature of the hyperbola or offers direct geometric supervision. This mismatch motivates the dedicated annotation tool proposed in this work, which allows annotators to place and adjust a parametric hyperbola band through a few interpretable controls, and to export both the shape parameters and the derived bounding box.

III. Proposed method

A. Hyperbola-Band Annotation

We designed HBAT, a visual annotation tool for hyperbola-band targets that supports parametric hyperbola annotation with automatic bounding-box generation. Its user interface is illustrated in Fig. 2.

First, the user selects an image folder, and the tool automatically loads all images within it. The original image is displayed on the lower-left panel, while the annotated image is shown on the right. A mouse click on the original image marks the vertex position of the hyperbola, and five adjustable parameters are provided on the right panel.

The five parameters are defined as follows: the horizontal coordinate x_{vertex} and vertical coordinate y_{vertex} of the hyperbola vertex, the slope s, the horizontal span, and the ribbon thickness. Among these, the slope s is the only parameter that governs the shape of the curve. Once the vertex is placed, a single slope value fully fixes the openness of the two arms, so a larger slope yields a narrower and sharper hyperbola and a smaller slope yields a flatter one. The centreline of the downward-opening band is

$$y_c(x) = y_{\text{vertex}}\sqrt{1+\left(\frac{s\,(x-x_{\text{vertex}})}{y_{\text{vertex}}}\right)^2}, \quad (1)$$

and the band is the vertical ribbon of all points (x, y) satisfying $|y - y_c(x)| \leq \text{thickness}/2$, where the span sets the horizontal extent over which the two arms are drawn. By clicking the Add as New Annotation button, the current parameter configuration is saved as a new annotation for the active image, and multiple annotations can be added to a single image.

All annotations are stored in JSON format with the following structure:

{

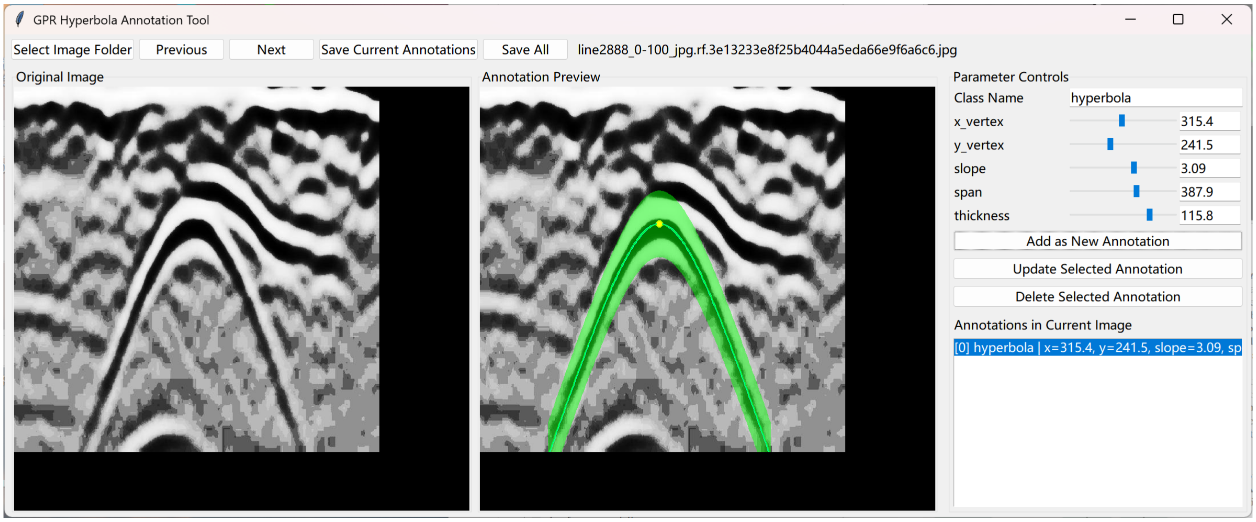
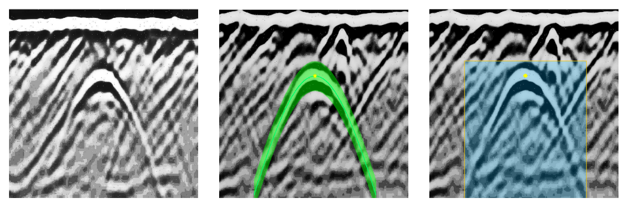


Fig. 2. The HBAT annotation interface

”line2888_0-100_jpg.rf.ce38dc3799
2454823c8dea9e0f814ac5.jpg”: [
{
”label”: ”hyperbola”,
”x_vertex”: 315.4,
”y_vertex”: 241.5,
”slope”: 3.09,
”span”: 387.9,
”thickness”: 115.8
}]
}

Here, the top-level key denotes the image filename, and each list entry stores one annotated hyperbola band as its parameters and their corresponding values.

After the hyperbola band is annotated, the conventional bounding-box annotation can be directly derived from the boundary of the band. Fig. 3 shows an example, presenting the raw image, the annotated hyperbola band, and the corresponding bounding-box form.



(a) Raw image. (b) Annotated hyperbola band. (c) Derived bounding box.

Fig. 3. Example of a hyperbola sample: the raw image, the annotated hyperbola band, and the bounding box derived from the band boundary.

B. Hyperbola-attention neural network

The network takes a single-channel grayscale B-scan image as input and outputs a rectangular bounding box for each hyperbolic signature in the image. Unlike existing methods, we introduce within the backbone an attention branch that is explicitly supervised by a hyperbola-band mask, and inject it into the bottleneck features via channel-wise concatenation, thereby guiding the network to focus on the hyperbolas and improving detection performance. A schematic of the network is shown in Fig. 4.

The backbone consists of three down-sampling stages followed by a bottleneck layer; each stage comprises two 3×3 convolutions, each followed by batch normalization and ReLU, together with a subsequent 2×2 max-pooling operation that doubles the channel depth while halving the spatial resolution. Starting from the 1×640^2 input, the three stages produce feature maps f_1 of size 32×320^2, f_2 of size 64×160^2, and f_3 of size 128×160^2, which are further reduced to the bottleneck feature F_b of size 256×80^2. The attention branch takes f_3 as input rather than the bottleneck feature F_b, as f_3 retains a higher spatial resolution that is better suited to precise localization.

The supervised attention branch is the central component of the proposed method. A lightweight head $g_a(\cdot)$, consisting of a 3×3 convolution with batch normalization and ReLU followed by a 1×1 convolution, maps f_3 to a single-channel logit that is passed through a sigmoid to yield the attention map $A \in [0, 1]^{H \times W}$, whose entry A_i is the predicted

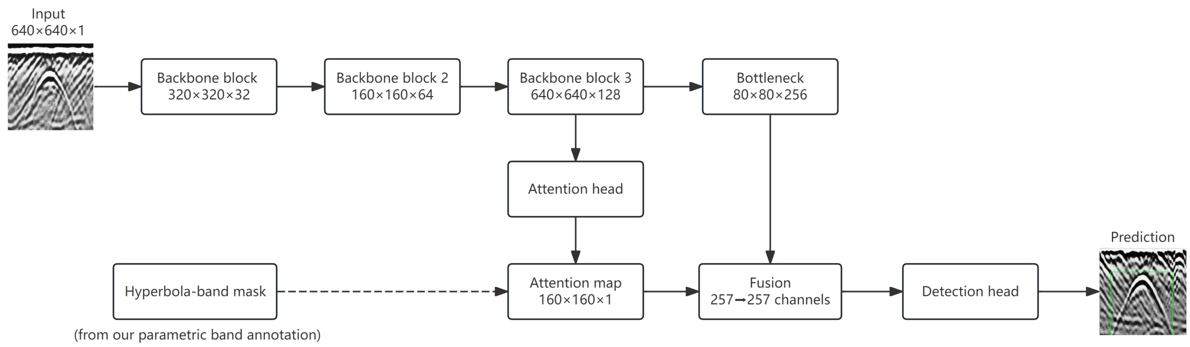


Fig. 4. Overview of the proposed network.

probability that pixel i belongs to the hyperbola band. The supervisory target $B \in \{0,1\}^{H\times W}$ is a binary band mask rasterized from the annotated hyperbola geometry at the same resolution. The attention map A is then driven to approximate B by a loss that combines a binary cross-entropy term and a Dice term, both computed between A and B over the $N = HW$ pixels,

$$\mathcal{L}_{\text{att}} = \mathcal{L}_{\text{bce}} + \mathcal{L}_{\text{dice}}, \tag{2}$$

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N}\sum_{i=1}^{N}\Big[B_i \log A_i + (1-B_i)\log(1-A_i)\Big], \tag{3}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_{i=1}^{N} A_i B_i}{\sum_{i=1}^{N} A_i + \sum_{i=1}^{N} B_i + \epsilon}, \tag{4}$$

where $\epsilon = 10^{-6}$ is a small constant added to the denominator to avoid division by zero. The cross-entropy term performs an independent per-pixel classification, penalizing each pixel that is misclassified as belonging to, or not belonging to, the hyperbola band. The Dice term instead measures the overall overlap between A and B, with the loss decreasing as the overlap increases. The two terms are complementary: the former enforces pixel-level accuracy, while the latter enforces region-level shape consistency and helps mitigate the class imbalance caused by the band occupying only a small fraction of the image.

a) Attention-guided feature fusion: The attention map A is passed through a sigmoid and downsampled by a factor of two via average pooling to align with the bottleneck resolution, yielding $\tilde{A}$. It is then concatenated with F_b along the channel dimension and compressed back to 256 channels by a 1×1 convolution with batch normalization and ReLU, producing the fused feature $\hat{F}$. This operation injects the spatial prior of the hyperbola band into the feature representation passed to the detection head.

b) Detection head and prediction: The detection head follows a YOLO-style dense prediction paradigm, producing three parallel outputs over an 80×80 grid at stride 8: objectness (1 channel), box size (2 channels), and sub-pixel center offset (2 channels). During inference, grid cells whose sigmoid-activated objectness exceeds a threshold τ are decoded into bounding boxes, and redundant detections are removed by box-level intersection-over-union (IoU)-based NMS. Let $\mathcal{P}$ denote the set of positive cells, namely the grid cells that contain an object center, and $\mathcal{N}$ the remaining background cells. The objectness map is supervised by a class-balanced binary cross-entropy over the hard targets p_j, which equal 1 at positive cells and 0 elsewhere,

$$\mathcal{L}_{\text{obj}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \text{BCE}(\hat{o}_j, 1) + \frac{\lambda_{\text{noobj}}}{|\mathcal{N}|}\sum_{j\in\mathcal{N}} \text{BCE}(\hat{o}_j, 0), \tag{5}$$

where $\hat{o}_j$ is the predicted objectness logit at cell j and $\lambda_{\text{noobj}} = 0.5$ down-weights the dominant background cells. The box-size and center-offset branches are regressed only at the positive cells with an ℓ_1 loss,

$$\mathcal{L}_{\text{size}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \|\hat{s}_j - s_j\|_1, \qquad \mathcal{L}_{\text{off}} = \frac{1}{|\mathcal{P}|}\sum_{j\in\mathcal{P}} \|\hat{\delta}_j - \delta_j\|_1, \tag{6}$$

where s_j and δ_j are the ground-truth box size and sub-pixel center offset and $\hat{s}_j$, $\hat{\delta}_j$ their predictions. The detection loss is the sum of the three terms,

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{size}} + \mathcal{L}_{\text{off}}, \tag{7}$$

TABLE I
Training Settings

Item	Setting
Optimizer	Adam
Learning rate	5×10^{-4}
Batch size	8
Training epochs	100
Mixed precision	bfloat16

and the overall training objective combines it with the attention supervision loss introduced above,

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda\,\mathcal{L}_{\text{att}}, \tag{8}$$

where λ controls the relative weight of the attention supervision. All terms are optimized jointly in a single end-to-end training pass.

IV. Result and evaluation

A. Implementation Details

The dataset used in this study is a publicly available GPR dataset, comprising 318 B-scan images in total. For each experiment, the images are randomly split into a training set according to a specified ratio, with the remaining images used as the test set. All experiments are conducted under ten random seeds, and the reported results are averaged over the ten runs, with the corresponding standard deviations provided. For evaluation, a predicted box is counted as a true positive if its IoU with a ground-truth box is at least 0.5, and each ground-truth box can be matched at most once; precision, recall, and F1-score are then computed accordingly.

All experiments are implemented in PyTorch and run on a single NVIDIA GeForce RTX 5060 Ti GPU (8 GB) with an AMD Ryzen 7 9800X3D CPU. The main training settings are summarized in Table I.

B. Detection performance

Table II reports the detection performance under different training-data ratios. As can be observed, the proposed method consistently outperforms the baseline across all training-data ratios. The improvement in F1-score is most pronounced at a training-data ratio of 0.7, where our method achieves 0.84 compared with 0.68 for the baseline. The training stability is also notably enhanced, with the standard deviation decreasing from 0.21 to 0.03. This gain is primarily attributed to the improvement in recall, as the attention constraint guides the network to focus on the hyperbola-band regions and thereby reduces missed detections, which demonstrates the effectiveness of the proposed method.

Nevertheless, at a training-data ratio of 0.9, no significant improvement is observed and the stability is relatively poor. This is presumably due to the limited number of samples in the test set, which leads to unstable evaluation results.

It should be noted that the above results are obtained with $\lambda = 0.7$; a sensitivity analysis of the parameter λ is further presented in Section IV-E.

C. Visualization of attention maps

To visualize where the network attends, we adopt gradient-weighted class activation mapping (Grad-CAM) [41], a widely used method for interpreting the predictions of convolutional networks. Intuitively, it produces a heatmap over the input image in which brighter regions are the ones that contribute most to the network's decision, so we can directly see which parts of the image the model relies on when making a detection. Fig. 5 shows a representative sample, where we use Grad-CAM to visualize the regions that the network relies on. The top row corresponds to the baseline without attention supervision, and the bottom row to our method; the left column shows the detection results and the right column the corresponding Grad-CAM maps. For the baseline, the attention is mostly scattered over the background, so the model fails to focus on the hyperbola band and does not detect it. In contrast, with the explicit supervision on the hyperbola band, our method attends not only to a few background curves but also to the hyperbola band itself, clearly capturing the overall hyperbolic structure and thus producing a correct detection. This comparison demonstrates the effectiveness of the proposed attention supervision.

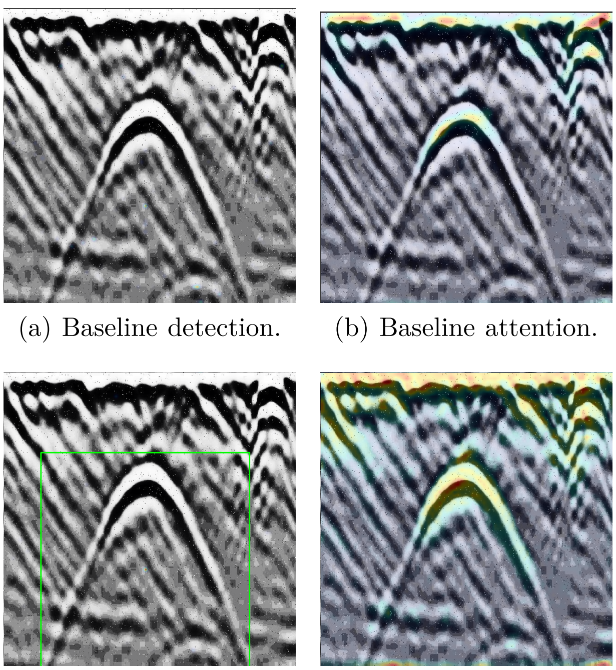
D. Comparison of Detection Performance with Different Attention Modules

To verify whether the proposed explicit geometric supervision is more effective than generic attention modules for hyperbola detection, we compare it with four representative attention modules [42]–[45] that have been applied to ground-penetrating radar, synthetic aperture radar (SAR), and remote-sensing recognition. The comparison results are summarized in Table III. For a fair comparison, we re-implement these attention modules at the same position within an identical backbone and evaluate them on our hyperbola dataset under exactly the same training and evaluation settings, so that all results reported in the table are obtained by our own re-implementation.

TABLE II
Detection performance (mean ± std over 10 seeds) at different training-data ratios. "Ours" uses the attention constraint with $\lambda_{\text{att}} = 0.7$. The best F1 per row is shown in bold.

Training ratio	Baseline			Ours		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0.5	0.81 ± 0.19	0.43 ± 0.09	0.53 ± 0.07	0.86 ± 0.08	0.49 ± 0.07	0.62 ± 0.06
0.6	0.71 ± 0.20	0.56 ± 0.08	0.60 ± 0.10	0.75 ± 0.24	0.65 ± 0.06	0.66 ± 0.12
0.7	0.75 ± 0.29	0.68 ± 0.10	0.68 ± 0.20	0.91 ± 0.08	0.78 ± 0.04	0.84 ± 0.03
0.8	0.84 ± 0.21	0.82 ± 0.07	0.81 ± 0.15	0.91 ± 0.12	0.88 ± 0.04	0.89 ± 0.07
0.9	0.87 ± 0.23	0.90 ± 0.06	0.87 ± 0.16	0.85 ± 0.21	0.93 ± 0.04	0.87 ± 0.16

Note: At ratio 0.9, Ours shows lower precision but higher recall than the baseline; these opposing shifts happen to offset, yielding the same F1 score.



(a) Baseline detection. (b) Baseline attention.

(c) Our detection. (d) Our attention.

Fig. 5. Visualization of attention maps.

TABLE III
Comparison of Detection Performance with Different Attention Modules

Method	Precision	Recall	F1-score
Pan et al. [42]	0.86 ± 0.20	0.82 ± 0.04	0.82 ± 0.11
N. Wang et al. [43]	0.75 ± 0.17	0.82 ± 0.03	0.77 ± 0.10
Z. Wang et al. [44]	0.84 ± 0.28	0.74 ± 0.23	0.78 ± 0.25
Shen et al. [45]	0.80 ± 0.23	0.84 ± 0.09	0.80 ± 0.17
Ours	$\mathbf{0.91 \pm 0.12}$	$\mathbf{0.88 \pm 0.04}$	$\mathbf{0.89 \pm 0.07}$

The proposed method achieves the best precision, recall, and F1, reaching an F1 of 0.89 with an improvement of about 9% over the second-best method. Our method attains the highest precision and recall simultaneously, together with the lowest F1 standard deviation, thus offering both higher detection accuracy and greater stability. This can be attributed to the fact that these attention modules merely re-weight channel or spatial features according to the features themselves, without explicit guidance on where to attend; in contrast, our method imposes explicit supervision on the attention branch through the hyperbola-band mask, directly guiding the network to focus on the hyperbolic target regions and thereby outperforming these generic attention modules.

E. Sensitivity analysis of the parameter λ

To investigate the sensitivity of the proposed method to the attention-constraint weight λ, we vary λ from 0.1 to 10 and evaluate the detection F1 over 10 random seeds for each value, where $\lambda = 0$ corresponds to the baseline. The resulting distribution of the F1 is shown in Fig. 6. Across the entire range of λ, the median F1 obtained with the attention constraint consistently remains between 0.78 and 0.84, substantially higher than the baseline value of 0.68. This indicates that the proposed method is insensitive to the choice of λ and yields stable performance gains over a wide range of this parameter. The best result is achieved at $\lambda = 0.7$, which attains the highest median F1 of 0.84 together with the most concentrated distribution, with a standard deviation of only 0.03. When λ is too small, the constraint becomes too weak to provide sufficient guidance, whereas an excessively large λ slightly suppresses the primary detection task and lowers the F1 to around 0.79. These results suggest the existence of an intermediate optimal range, and $\lambda = 0.7$ is therefore adopted as the default setting throughout our experiments in consideration of both accuracy and stability.

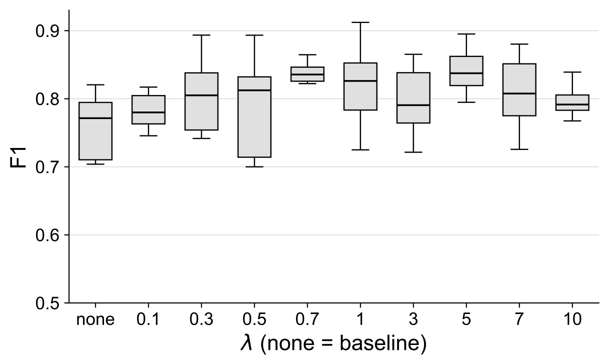


Fig. 6. Sensitivity of the detection F1 to the attention-constraint weight λ.

TABLE IV
Detection performance on an additional GPR dataset.

Method	Precision	Recall	F1-score
Baseline	0.68 ± 0.08	0.83 ± 0.04	0.75 ± 0.05
Ours	$\mathbf{0.76 \pm 0.06}$	0.83 ± 0.06	$\mathbf{0.79 \pm 0.04}$

F. Generalization to an Additional Dataset

We retrain and evaluate our method on an additional public GPR dataset comprising 247 B-scan images, following the same training and evaluation protocol described in Section IV-A. Table IV reports the resulting precision, recall, and F1-score. The proposed method again outperforms the baseline, improving the F1-score from 0.75 to 0.79 and the precision from 0.68 to 0.76, while recall remains essentially unchanged. This indicates that the benefit of explicit geometric attention supervision is not limited to the primary dataset and consistently holds when the method is trained on a different GPR dataset.

V. Discussion

A. On the choice of output representation

Although our annotation tool provides a complete parametric description of each hyperbola, in this work we do not train the network to regress these parameters directly, for two main reasons. First, the mainstream detection task, together with its datasets and evaluation protocols (IoU-based precision, recall, and F1), is built around bounding boxes; retaining a bounding-box output therefore allows a fair comparison with mainstream detectors and generic attention modules under a fully identical protocol. Second, directly regressing hyperbola parameters such as slope is sensitive to annotation noise, clutter, side lobes, and mutually overlapping echoes, while the disparate scales of the individual parameters make the training loss difficult to balance.

B. Cross-dataset domain shift

The generalization experiment in Section IV-F retrains the network from scratch on the additional dataset, which confirms that the proposed attention supervision is beneficial regardless of which dataset it is trained on, but it does not probe how well a model trained on one dataset performs on the other without retraining. To examine this more demanding setting, we train the network on one of the two datasets at a training ratio of 0.7 and, without any fine-tuning, directly evaluate the resulting model on the other; this is repeated in both directions, and all numbers below are averaged over ten random seeds.

Table V reports the results, reusing the in-domain F1 values already established at this training ratio in Table II (primary dataset) and Table IV (additional dataset). Cross-dataset performance collapses sharply in both directions, with F1 falling to roughly 0.08–0.16. The two datasets likely differ in more than the shape of the hyperbolas. Acquisition equipment, subsurface conditions, and general B-scan appearance may all differ between them, so neither model has learned anything that carries over to the other.

Such cross-dataset degradation is a well-documented difficulty in GPR interpretation more broadly, and is not unique to hyperbola detection: it has motivated dedicated domain-adaptation techniques for GPR data inversion [46] and for GPR object detection itself [47], as well as a range of transfer-learning strategies for related GPR recognition tasks [48]–[50]. Our approach does not adopt any of these dedicated domain-adaptation techniques; the improvement observed above comes only from the explicit attention supervision described earlier.

C. Future work

Although directly regressing hyperbola parameters is challenging for the reasons discussed above, it remains the most informative output for practical use, and overcoming this difficulty is a natural direction for future work: moving from bounding-box localization toward predicting the parametric hyperbola itself and exporting its parameters for inversion. In GPR interpretation, inversion aims to recover the physical properties of a buried target from its hyperbolic signature: the apex position and the opening of the curve are governed by the target depth and the propagation

TABLE V
Cross-dataset detection performance.

Train→Test	Method	In-domain F1-score	Cross Precision	Cross Recall	Cross F1-score
Primary→Additional	Baseline	0.68 ± 0.20	0.42 ± 0.07	0.05 ± 0.02	0.08 ± 0.02
	Ours	0.84 ± 0.03	$\mathbf{0.42 \pm 0.06}$	$\mathbf{0.05 \pm 0.02}$	$\mathbf{0.09 \pm 0.04}$
Additional→Primary	Baseline	0.75 ± 0.05	0.14 ± 0.04	0.14 ± 0.04	0.14 ± 0.03
	Ours	0.79 ± 0.04	$\mathbf{0.17 \pm 0.05}$	$\mathbf{0.17 \pm 0.05}$	$\mathbf{0.16 \pm 0.03}$

velocity in the host medium, so that accurate parameters translate into estimates of burial depth, object radius, and the dielectric permittivity of the surrounding soil, which are the quantities of practical interest for utility mapping and condition assessment [27]. Several recent studies already predict such hyperbola parameters directly instead of boxes, for example through transformer-based regression [30], [51] or vertex-oriented keypoint and state-space formulations [14], [39]. The quantities we would ultimately estimate, namely the apex position and the opening of the curve, are essentially the same as those targeted by these works, so we do not claim a new set of parameters to predict. What differs is our starting point: we already have a complete parametric annotation of each hyperbola band, so moving on to predict these parameters is a natural next step for us. In addition, our attention branch is already supervised by the hyperbola-band mask to produce a band-shaped response that closely follows the target geometry, which offers a natural bridge toward such parametric prediction. A promising direction is therefore to attach a lightweight regression head on top of this response to decode the vertex, slope, span, and thickness of each hyperbola, rather than building a separate parameter regressor from scratch, so that the model outputs the complete parametric hyperbola end to end and directly supports downstream inversion.

VI. Conclusion

In this paper, we proposed a hyperbola-band attention method to improve hyperbola detection in GPR B-scan images. We first developed a dedicated annotation tool that describes each hyperbola as a parametric band through a few interpretable controls and exports both the shape parameters and the derived bounding box. Building on a standard bounding-box detector, we introduced an attention branch explicitly supervised by the hyperbola-band mask, injecting the geometric prior into the feature representation and thereby improving detection performance. Experiments on a public GPR dataset show that the proposed supervision consistently improves detection across training-data ratios. At a training-data ratio of 0.7, it raises the F1-score from 0.68 to 0.84, with the gain driven mainly by higher recall. The standard deviation across random seeds is also reduced from 0.21 to 0.03, indicating that the geometric supervision stabilizes training and reduces sensitivity to initialization, which is particularly valuable under limited training data. The method also outperforms other attention mechanisms, confirming its effectiveness. While the current work focuses on localizing and classifying hyperbolas, future work will extend the framework to infer the hyperbola's shape parameters directly, meeting the requirements of downstream tasks such as geophysical inversion.

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